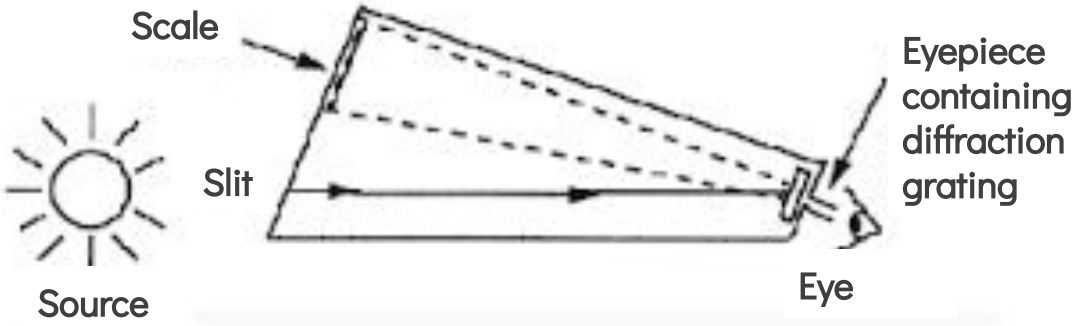
Activity: Spectroscopy

Looking at Light!

Activity - Spectroscopy

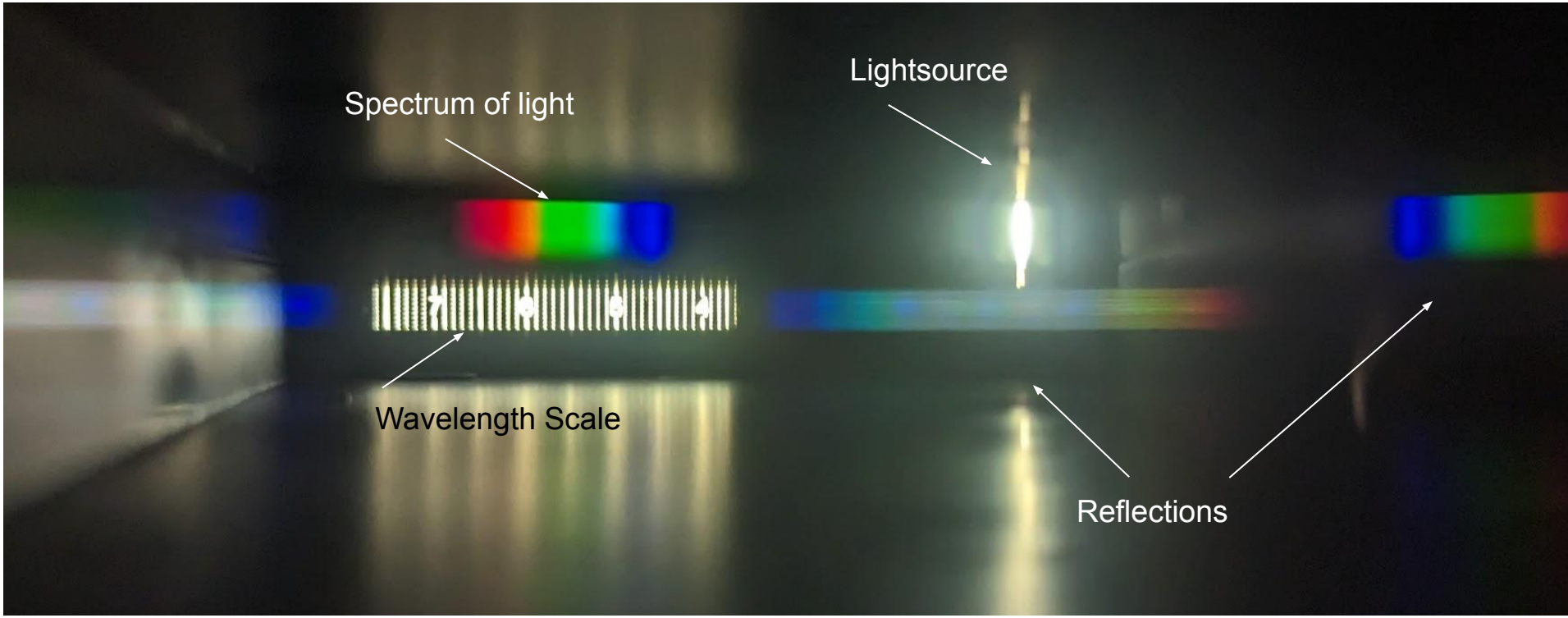
A simplified diagram of the spectroscope you will use today and how it works is shown below. The slit regulates the amount of light which enters the spectroscope and thus the width of the resulting spectral lines. Light entering the slit passes through the spectroscope to the diffraction grating located in the eyepiece. Light passing through this grating is separated into its components. The scale, which is visible through the eyepiece, is illuminated by the light entering the spectroscope from a second direction. Thus, you will see the spectrum superimposed upon the scale, which is marked off in units of angstroms.



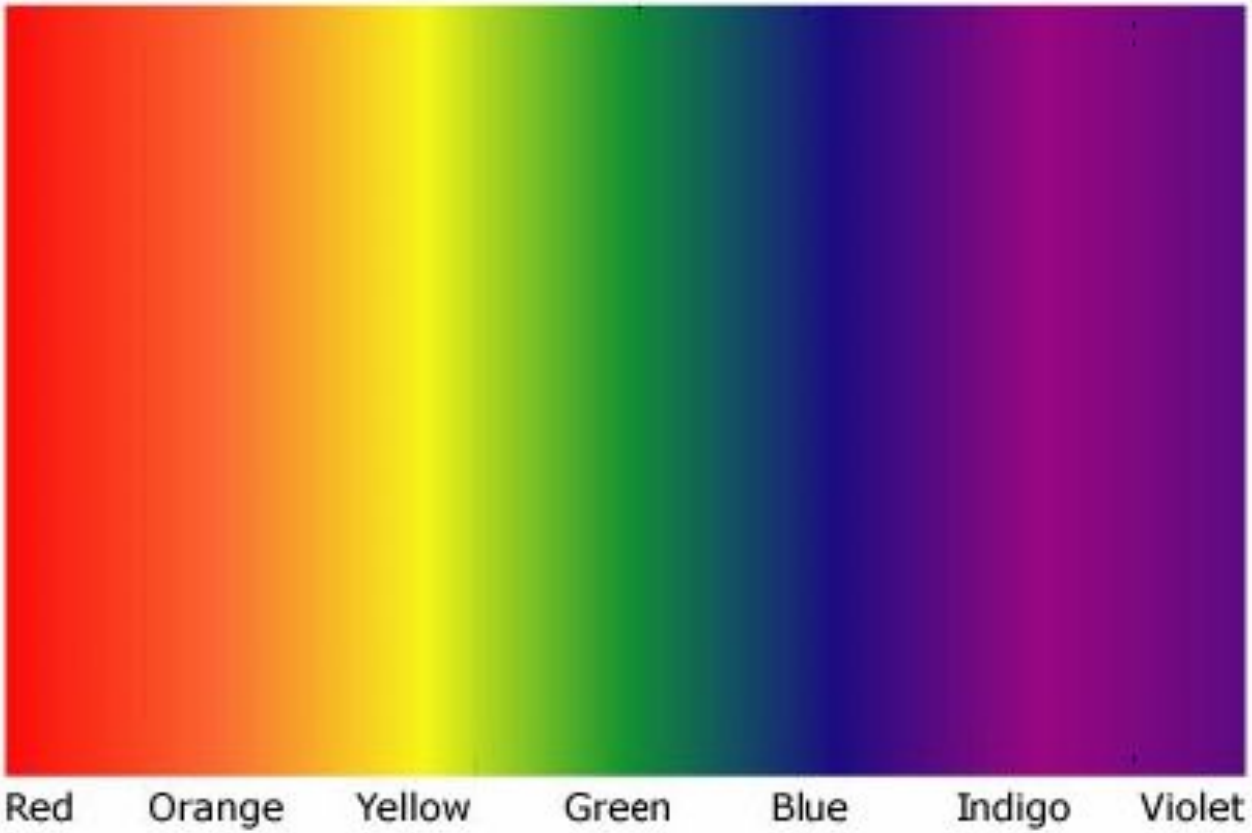
Procedure

1. Look through the eyepiece of a spectroscope at a light source or gas discharge tube. Make sure the slit faces the light and that it is parallel to the light source.
2. Look for a spectrum inside the spectroscope to the side of the slit. You might have to move the spectroscope slightly to the side of the light source or gas discharge tube to see the spectrum.
3. Draw each spectrum that you observe. Use coloured pencils.

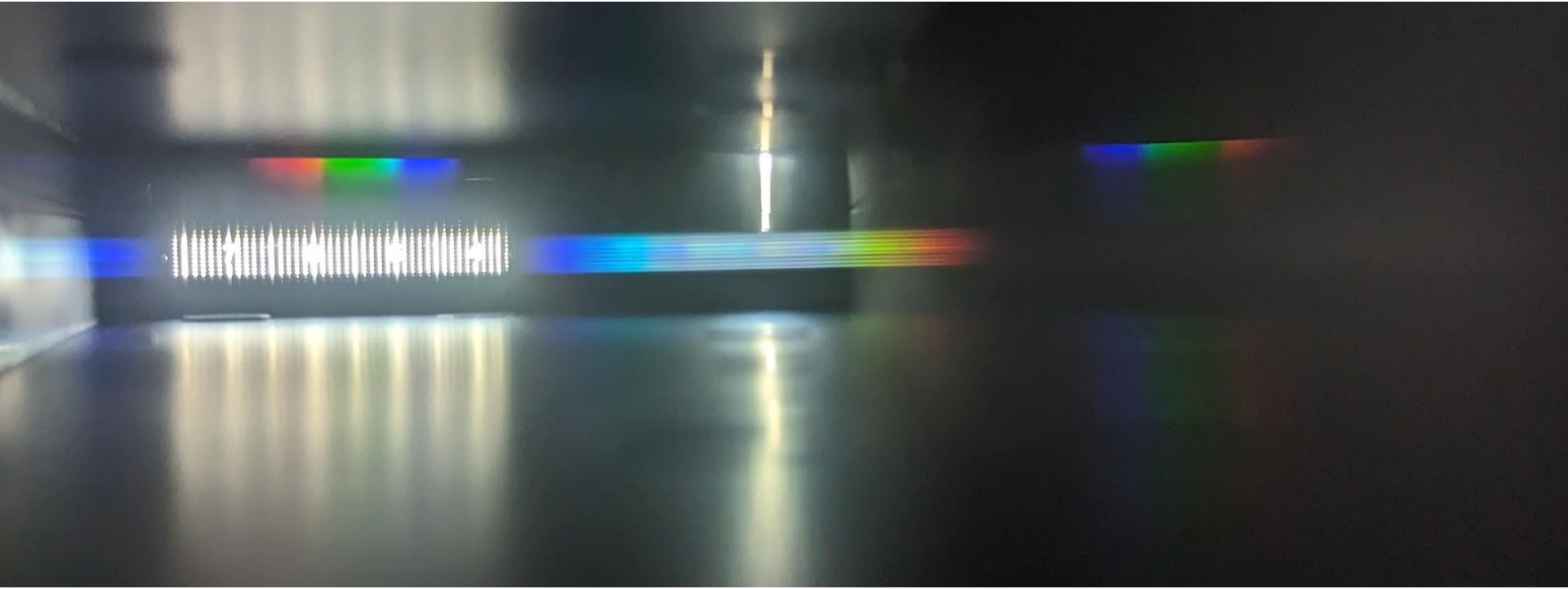
Inside a spectroscope!



Sunlight

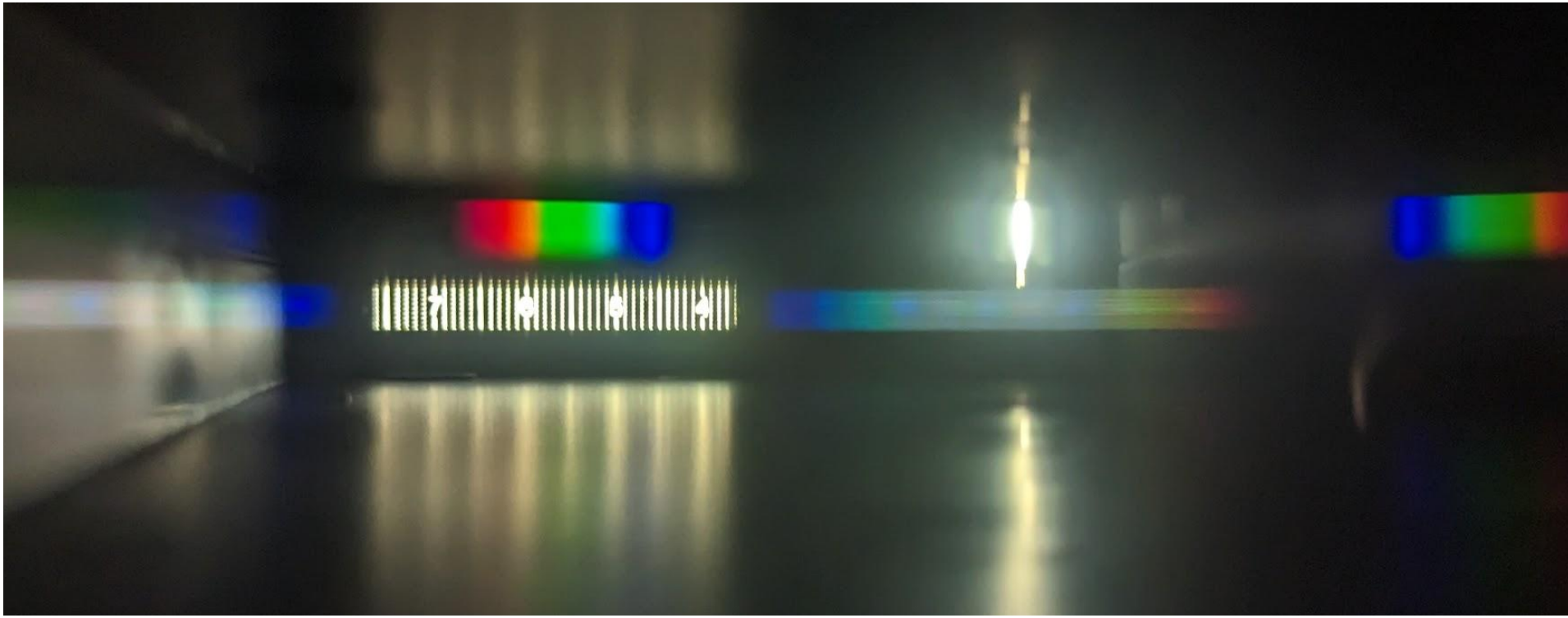


Sunlight

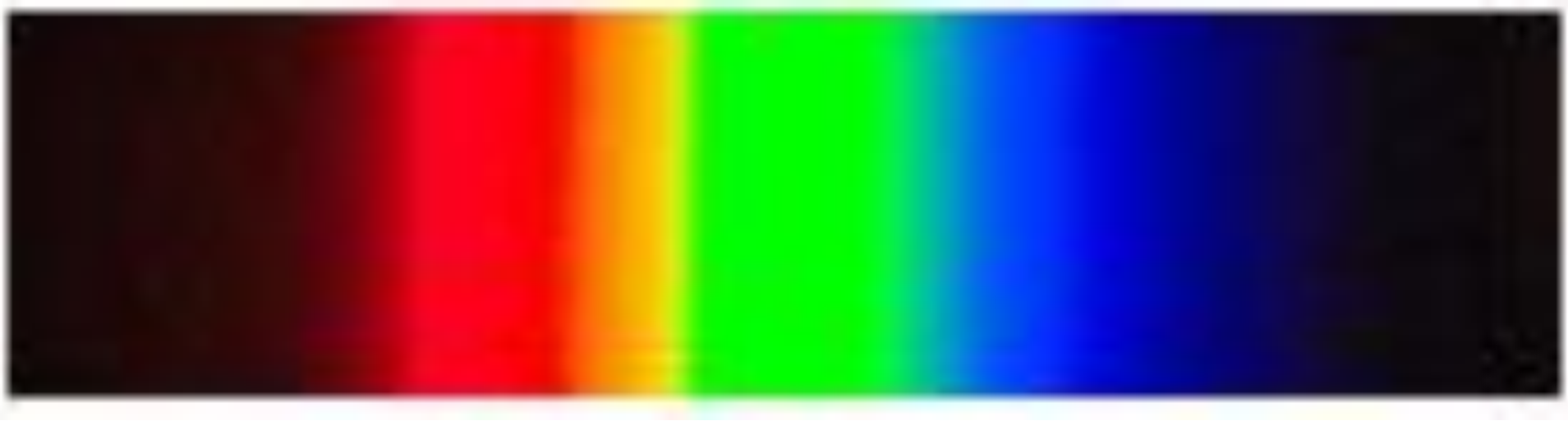


LED Light

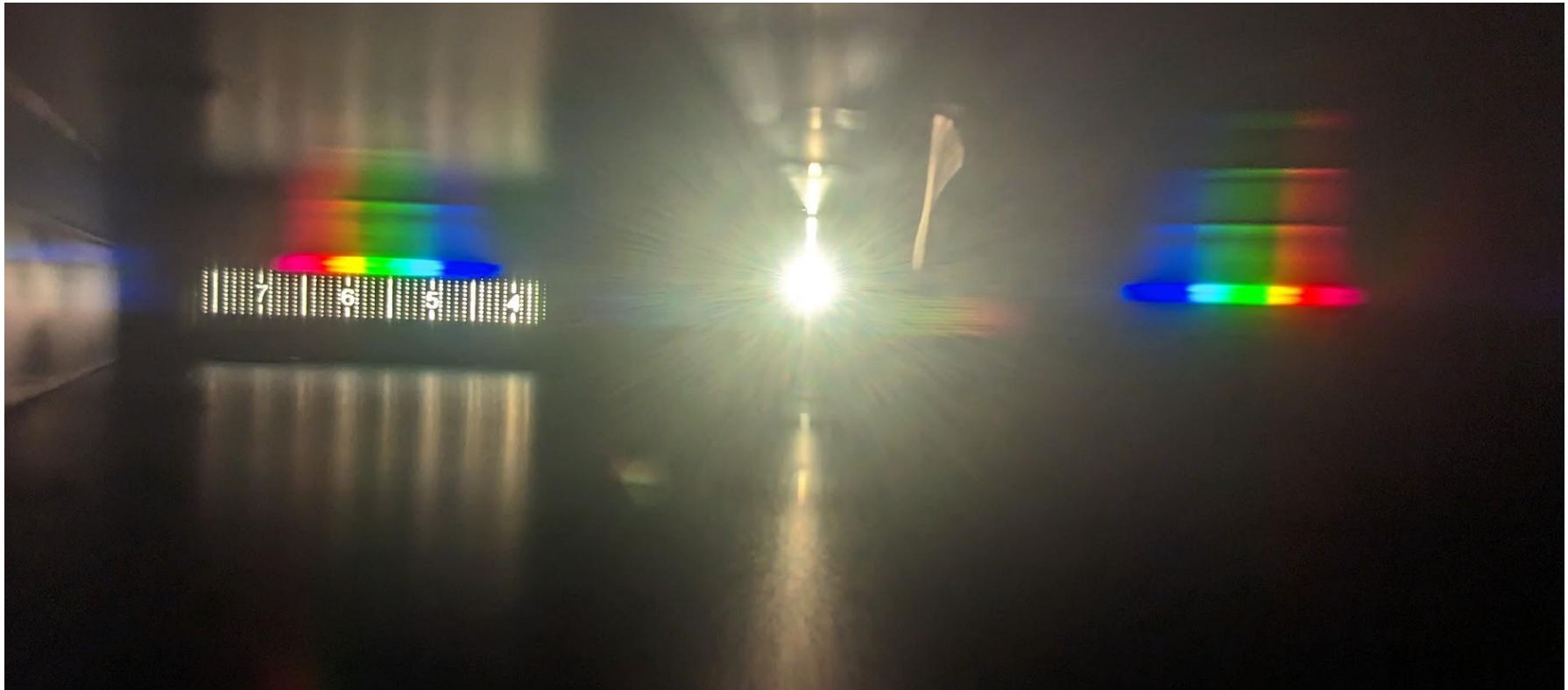
LED Light



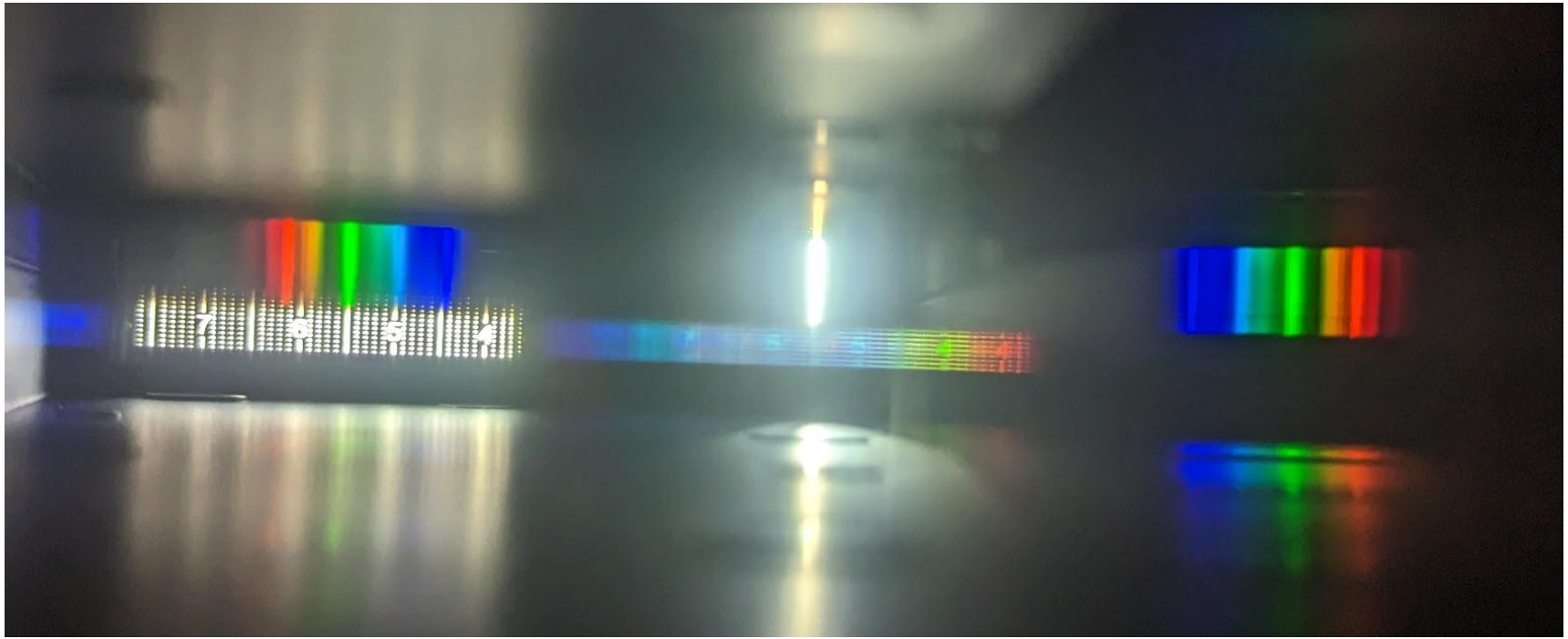
Incandescent Light



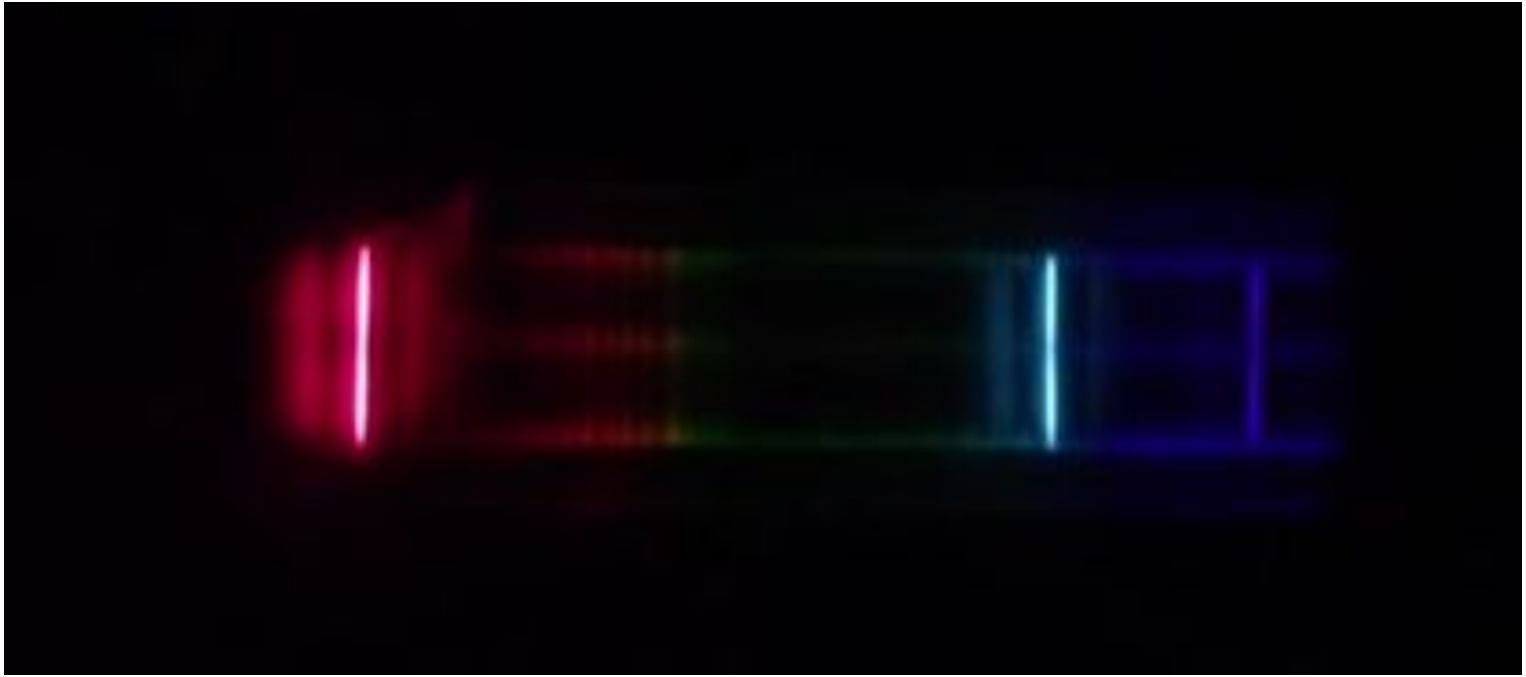
Incandescent Light

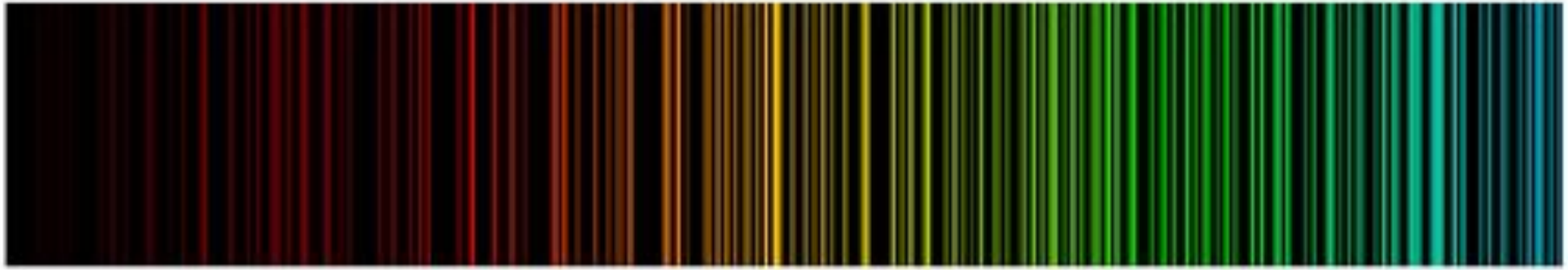


Fluorescent Light



Hydrogen



Neon

Argon